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Practical Work Project

Module: Image Processing

Project Title:

Image Captioning Using Retrieval-Augmented Generation (RAG)
phase 1

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Chapter 1

Introduction

Background

Image processing projects rely on the quality and organization of the data used throughout their development. Working with raw image datasets may involve several challenges, such as missing files, duplicate images, corrupted data, and inconsistencies between images and their corresponding descriptions.

For this reason, data exploration and preprocessing are essential steps before performing further experiments. They help us understand the structure of the dataset, identify potential problems, organize the available information, and ensure that the data is suitable for subsequent processing.

In this project, the COCO (Common Objects in Context) dataset is used as the main source of images and their associated captions. It provides images accompanied by textual descriptions, making it suitable for studying the relationship between visual content and natural language.

1.1 Objective of Phase 1

The main objective of Phase 1 is to explore, organize, and prepare the selected COCO data. This phase focuses on understanding the dataset structure, checking the integrity of images and captions, and applying the necessary preprocessing steps.

Another important objective is to ensure that the selected data is properly organized and that the relationship between each image and its corresponding captions is preserved. Quality control checks are also considered to detect possible errors and maintain the consistency of the prepared dataset.

1.2 Scope of the Work

Phase 1 covers several tasks related to data exploration and preprocessing. These tasks include:

- **Dataset Exploration:** examining the COCO dataset structure, its images, captions, and available annotations.
- **Data Selection and Organization:** selecting the required data and organizing the image IDs, captions, and dataset splits.
- **Image Preprocessing:** preparing images in a consistent format for further processing.
- **Caption Preprocessing:** examining and preparing textual descriptions while preserving their relationship with the original images.

- **Metadata Preparation:** organizing relevant information associated with images and captions.
- **Quality Control:** performing checks to identify missing data, duplicates, inconsistencies, and other potential issues.

This phase establishes the foundation for the next stages of the project by providing a structured and carefully examined dataset.

Chapter 2

Preprocessing Steps

2.1 Introduction to the COCO Dataset

The Common Objects in Context (COCO) dataset is a large-scale dataset commonly used in computer vision research. It contains images representing everyday scenes, together with textual descriptions (captions) and object annotations.

In this project, we use the COCO 2017 dataset as the main source of visual and textual data. Each image is associated with a unique identifier (Image ID), which allows it to be linked to its corresponding captions and annotations.

2.2 Dataset Structure

The COCO dataset provides several types of information:

- **Images:** Photographs representing different real-world scenes and objects.
- **Captions:** Natural language descriptions explaining the visual content of each image.
- **Image IDs:** Unique identifiers used to distinguish images and link them to their annotations.
- **Object Annotations:** Information describing the objects present in images, including their categories.

2.3 Selected Dataset Subset

After exploring the original dataset, a subset of 10,000 unique images was selected from the COCO 2017 training set.

Each selected image was associated with exactly five captions, resulting in a total of 50,000 captions.

The selected subset was then examined to verify the consistency of image IDs and caption associations. The audit confirmed that all 10,000 selected images had exactly five captions, with no missing captions or duplicate image IDs.

2.4 Data Audit and Validation

Data auditing and validation are essential steps in preparing a reliable dataset for the image captioning and Retrieval-Augmented Generation (RAG) pipeline.

The main objective is to verify the quality, consistency, and integrity of the selected COCO dataset before using it to build the knowledge base. This process helps identify missing data, corrupted images, duplicate records, and inconsistencies between images and their corresponding captions.

In our project, the audit focuses on ensuring that the selected images and their annotations are valid and correctly organized.

Ensure that each caption is associated with the correct image ID.
he audit was applied to the three splits. Each image file was opened and verified, and an MD5 fingerprint of each file was compared with the others to detect exact duplicates. No image was missing or corrupt. One pair of knowledge-base images (IDs 531721 and 438867) was found to be the same file stored under two image IDs. This pair is documented in the audit report, and no image has been removed so far. The results are saved in `audit_report.json` and `audit_documentation.json`.

These checks help prevent errors from propagating into the embedding generation and retrieval stages.

2.5 Image Preprocessing

Image preprocessing prepares the COCO images for the visual encoder. The 10,000 knowledge-base images were first inspected: 9,979 are RGB images and 21 are grayscale images with a single channel, none requires an EXIF orientation correction, and their sizes range from 72×51 to 640×640 pixels. Since the encoder expects three-channel images of a fixed size, a common format is required.

2.6 Caption Preprocessing

The purpose of caption preprocessing is to prepare the captions for retrieval while keeping the original captions as the ground truth for evaluation. For every caption, two fields are stored: `caption_raw`, the original text, used only for evaluation, and `caption_clean`, used only for indexing and retrieval. he cleaning converts the text to lowercase, replaces punctuation marks with spaces and normalises whitespace. For example, the caption "Woman in swim suit holding parasol on sunny day." becomes "woman in swim suit holding parasol on sunny day". Both fields are saved for the 50,000 knowledge-base captions and for the 500 validation captions, while the test reference captions are never modified. A separate tokenisation step is not applied yet and will be added when lexical retrieval is implemented.

2.7 Metadata

In this step, we extracted metadata from the COCO instance annotations. For each image, we collected information such as its ID, dimensions, object categories, bounding boxes, and the number of objects. We extracted metadata for 10,100 images. We found that 84 images had no annotated objects, and the average number of objects per image was 7.22. This metadata provides structured information about the visual content of our dataset

2.8 Embedding

CLIP ViT-B/32 is used as the image encoder: each preprocessed image is converted into a 512-dimensional vector. Captions are encoded with the Sentence-BERT model `all-MiniLM-L6-v2`, which maps each cleaned caption to a 384-dimensional vector. All vectors are L2-normalised and saved in `.npy` format. The embedding matrices have shapes $(10000, 512)$ for the images and $(50000, 384)$ for the captions. Computation was performed on a CPU and took about 35 minutes for the images and 7 minutes for the captions

2.9 Knowledge-Base Construction

The objective of knowledge-base construction is to organize the preprocessed COCO training data and its embeddings into a searchable structure. This knowledge base provides the external information required by the Retrieval-Augmented Generation (RAG) pipeline.

2.9.1 Vector Index Construction

FAISS is used to build two indexes: one for the 10,000 image vectors (dimension 512) and one for the 50,000 caption vectors (dimension 384). Both use the exact index `IndexFlatIP`. Since the vectors are L2-normalised, the inner product equals the cosine similarity. Each vector is linked to its source by its row position: row i corresponds to the i-th image ID, or to the i-th caption of the processed caption file.

2.9.2 Role in the RAG Pipeline

During retrieval, the embedding of a query image is compared with the vectors stored in the knowledge base. The most relevant results are selected and their associated captions can be used as contextual information for the caption-generation stage.

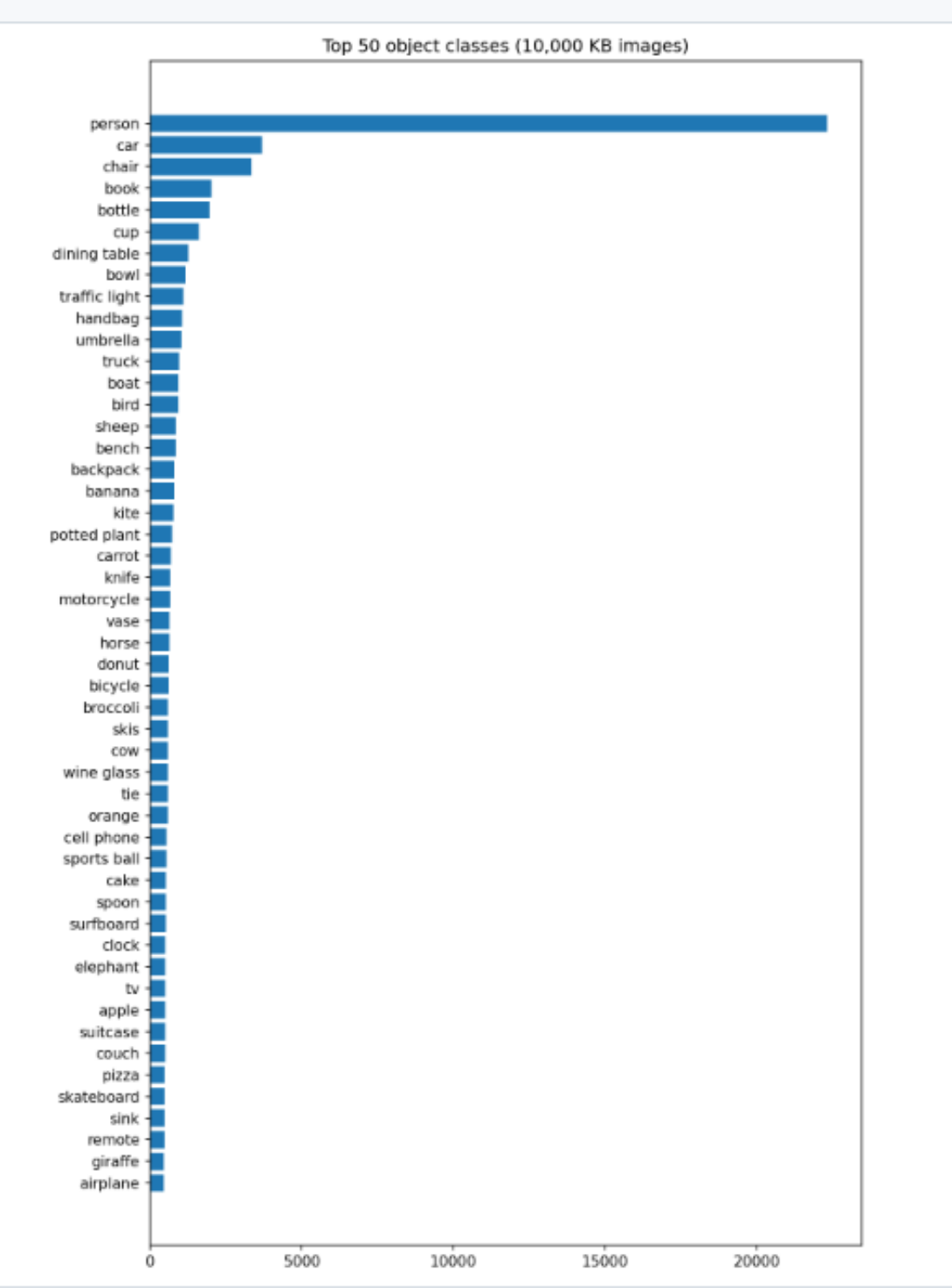


Figure 2.1: Top 50 object classes

Description :

This horizontal bar chart illustrates the frequency distribution of the top 50 object classes detected across a dataset of 10,000 KB images Dominant Category: The "person" class significantly outperforms all other object classes, appearing over 22,000 times Secondary Categories: Everyday items and common transport objects such as "car", "chair", "book", "bottle", and "cup" form the next highest detected classes, though at a much lower frequency compared to "person

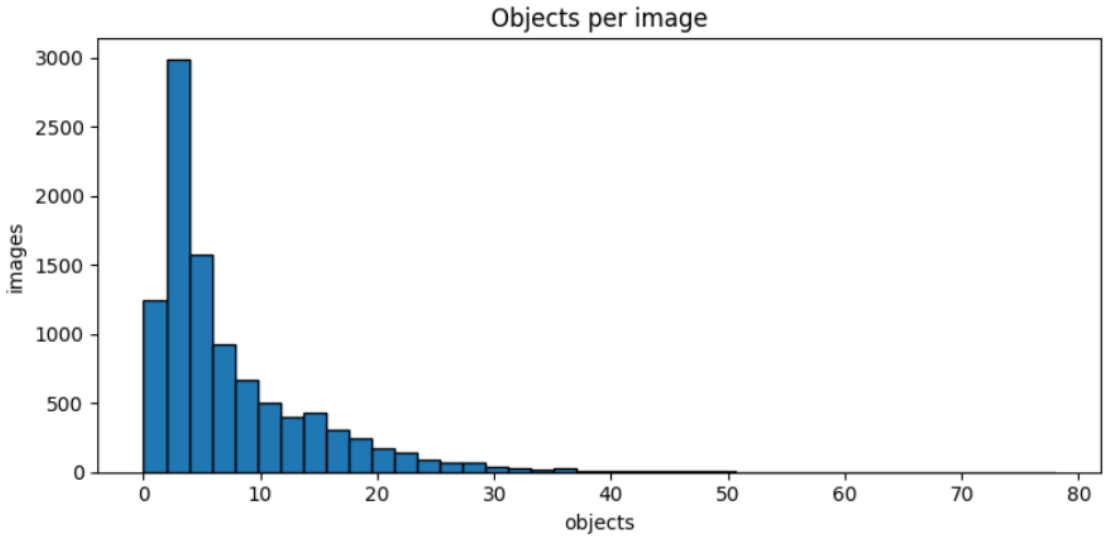


Figure 2.2: Objects per image

Description :

This histogram illustrates the distribution of the number of objects per image. The majority of images contain a small number of objects, with a sharp peak appearing between 2 and 4 objects. The frequency then decreases steadily as the complexity and number of objects per image increase

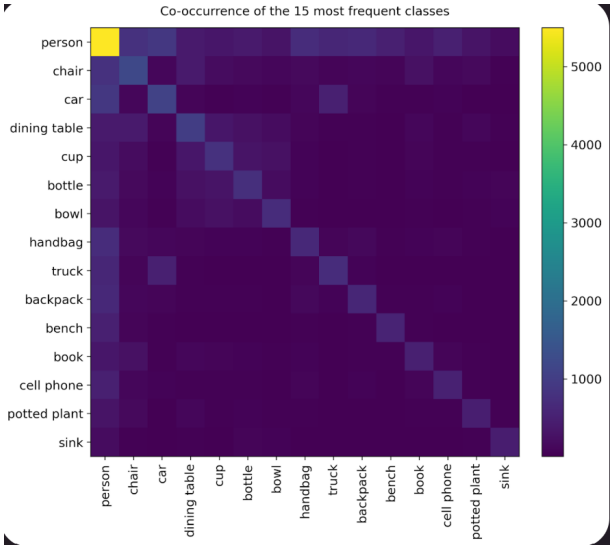


Figure 2.3: Co Occurences of the 15 most frequent classes

Description:

The strong diagonal line shows how often classes appear with themselves, while the brighter interactions highlight categories that frequently appear together in the same images, such as people with chairs or dining tables.

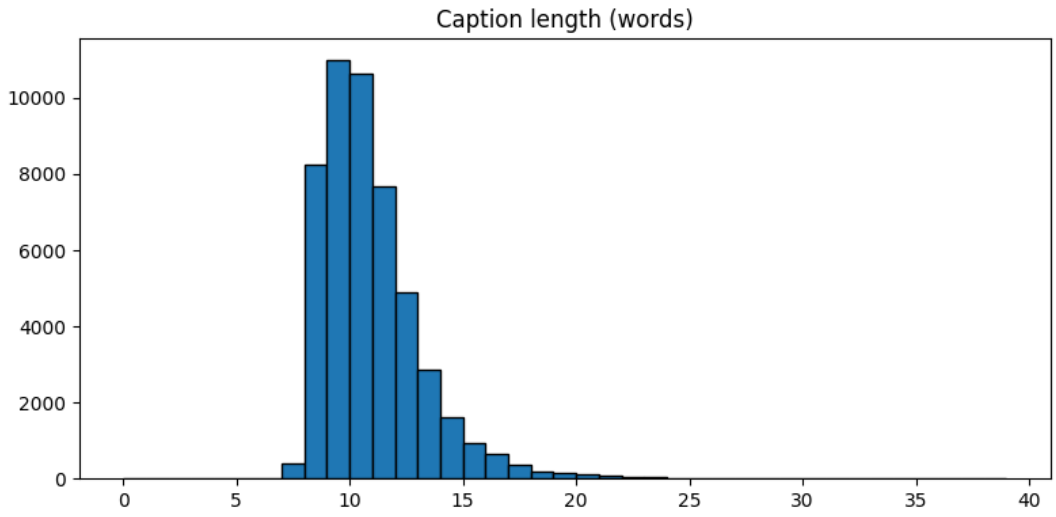


Figure 2.4: Caption Length

Description

This histogram illustrates the distribution of caption lengths measured in terms of word count The vast majority of captions contain between 8 and 13 words, with a prominent peak around 9 to 11 words The frequency decreases sharply beyond 15 words, showing that excessively long captions are rare

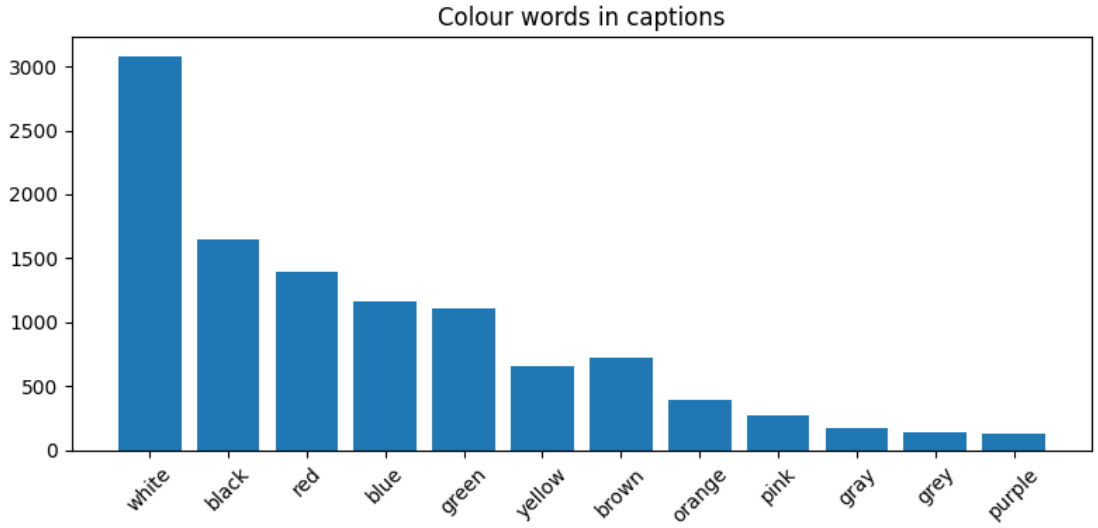


Figure 2.5: Colour words in Captions

Description

This bar chart shows the frequency of color words used in the image captions Primary colors like black, red, blue, and green are also frequently mentioned (between 1,000 and 1,700 occurrences) Colors such as pink, gray, grey, and purple appear least often, each with fewer than 300 instances

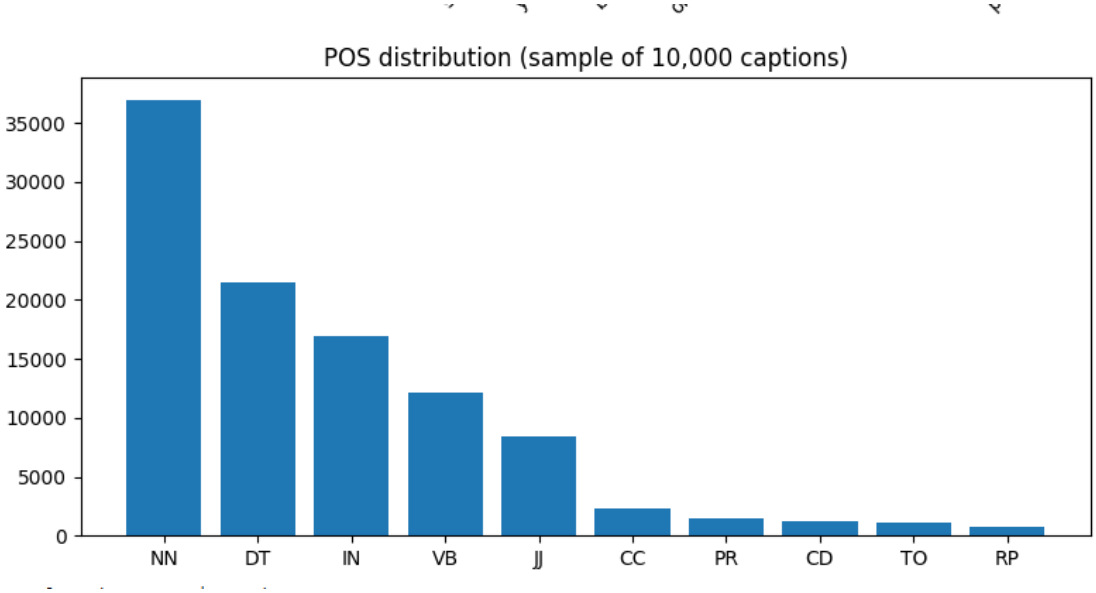


Figure 2.6: POS distribution

Description

This bar chart shows the part-of-speech (POS) distribution from a sample of 10,000 captions Nouns (NN) are the most frequent category, exceeding 35,000 occurrences Determiners (DT) and Prepositions (IN) follow as the next most common types Other grammatical categories like Verbs (VB) and Adjectives (JJ) also appear frequently, while categories such as conjunctions and particles are much less common